Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/24/2006
PI Name:	Hines, John M.S.		
Project Title:	Advanced Technology Development-Biosensors		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john.w.hines@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5538
Organization Name:	NASA Ames Research Center		
PI Address 1:	Mail Stop 302-306		
PI Address 2:			
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:	Co-I Karl Schweighofer (karl@mail.arc.nasa.gov) NASA Center for Astrobioinformatics		
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/2001	**End Date:**	10/01/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The technology scope for ATD-B is in the areas of advanced biomedical, biological,environmental and process control sensors; biotelemetry systems; embedded and in-situ monitoring and control instrumentation systems; smart data acquisition and management systems; and supporting technologies.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The NASA developed SHMS has application to medical care, DOD, Home Land security and a number of industry uses. -We have discussed the possibility of collaborating with the University of Miami Medical Center to utilize wireless communications in areas of medical research and patient physiological data collection. -The Astrobionics Office received funding from the Natick Soldier Center to develop a PAN for use on soldiers to monitor physiological status under extreme conditions. -We have demonstrated the SHMS as a usable tool that could transmit water quality data that could identify various biological and chemical contaminates in drinking water supplies. -We have explored the feasibility of developing a pH sensor that could be implanted in grazing animals (cattle and exotic zoo animals) that would be helpful in detecting the onset of rumen acidosis, which is a debilitating and often fatal disease.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)